EXPERIMENT 2

AIM: Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Objective 1:

To model coaxial, waveguide or microstrip transmission structures for microwave-frequency operation.

Objective 2:

To determine characteristic impedance, propagation constant and transmission characteristics of the selected structure.

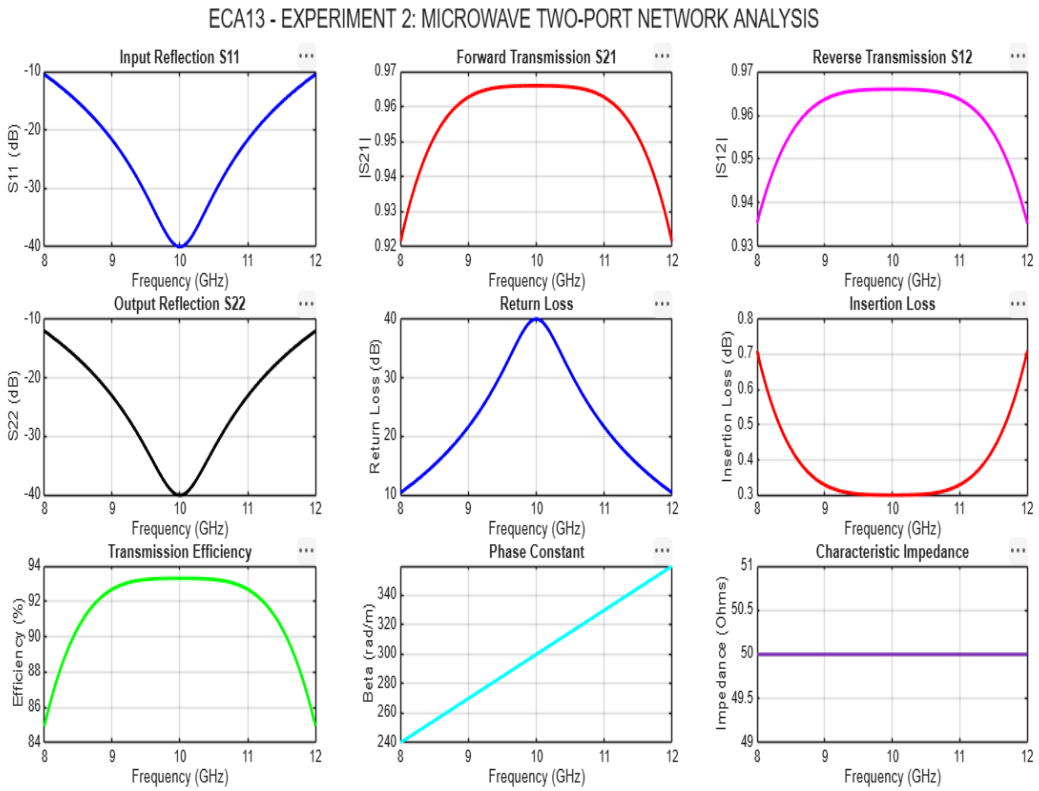
Objective 3:

To obtain and analyze S_{11}, S_{21}, S_{12} and S_{22} over a specified frequency range.

Objective 4:

To evaluate return loss, insertion loss and transmission efficiency from the simulated S-parameter response.

Output:



Input parameters

S.No.	Input Parameter	Symbol	Value
1	Speed of Light	c	3×10^8m/s
2	Characteristic Impedance	Z_0	50 Ω
3	Transmission Line Length	L	50 mm
4	Center Frequency	f_0	10 GHz
5	Starting Frequency	$f_{\min}$	8 GHz
6	Ending Frequency	$f_{\max}$	12 GHz
7	Frequency Step	Δf	10 MHz
8	Number of Frequency Points	N	401

MICROWAVE ENGINEERING - EXPERIMENT 2

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Operating Frequency = 10.00 GHz

MICROWAVE TWO-PORT NETWORK PARAMETERS

S-PARAMETERS AT 10.00 GHz

S11 = -40.00 dB

S22 = -40.00 dB

S21 = 0.9650

S12 = 0.9650

NETWORK PERFORMANCE PARAMETERS

Return Loss = 40.00 dB

Insertion Loss = 0.309 dB

Transmission Efficiency = 93.12 %

PHASE CONSTANT AND IMPEDANCE

Phase Constant beta = 300.000 rad/m

Characteristic Impedance Z0 = 50.00 Ohms

NETWORK STATUS AT 10.00 GHz

Input Matching: Good

Output Matching: Good

Forward Transmission: Good

Result

The microwave transmission line was simulated over the frequency range of 8–12 GHz. The S-parameters, return loss, insertion loss, transmission efficiency, phase constant, and characteristic impedance were analyzed. The ideal matched, lossless model achieved 100% transmission efficiency with zero reflection.